

Question ID: ba61d95f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Medium

Question

The population of Greenville increased by **7%** from **2015** to **2016**. If the **2016** population is ***k*** times the **2015** population, what is the value of ***k***?

Answer

- A. **0.07**
- B. **0.7**
- C. **1.07**
- D. **1.7**

Correct Answer: C

Rationale

Choice C is correct. Let ***x*** be the 2015 population of Greenville. It's given that the population increased by **7%** from 2015 to 2016. The increase in population can be written as **(0.07)*x***. The 2016 population of Greenville is given as the sum of the 2015 population of Greenville and the increase in population from 2015 to 2016. This can be rewritten as ***x* + (0.07)*x***, or **1.07*x***. Therefore, the value of ***k*** is **1.07**.

Choice A is incorrect. This is the percent, represented as a decimal, that the population increased from 2015 to 2016, not the value of ***k***.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of ***k*** if the population increased by **70%**, not **7%**, from 2015 to 2016.